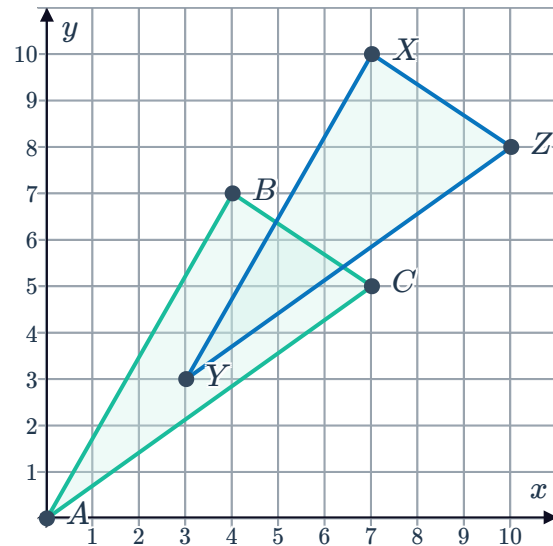


Coordinate methods and triangle congruence

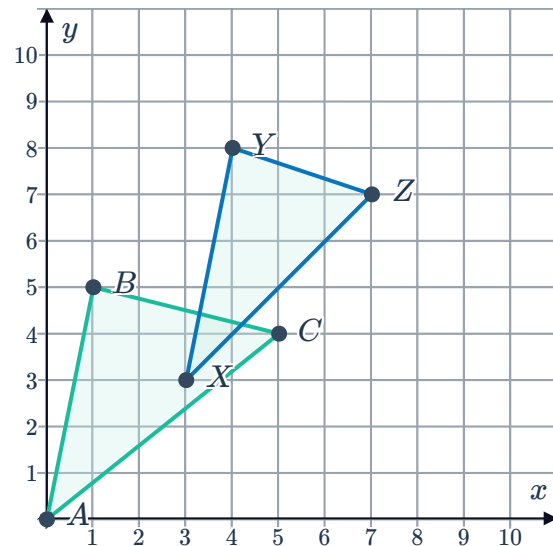
1 Consider the following diagram.

- a Describe the translations that must be applied to move X to B .
- b Hence, which points of $\triangle ABC$ will Y and Z translate to?
- c Are $\triangle ABC$ and $\triangle YXZ$ congruent?

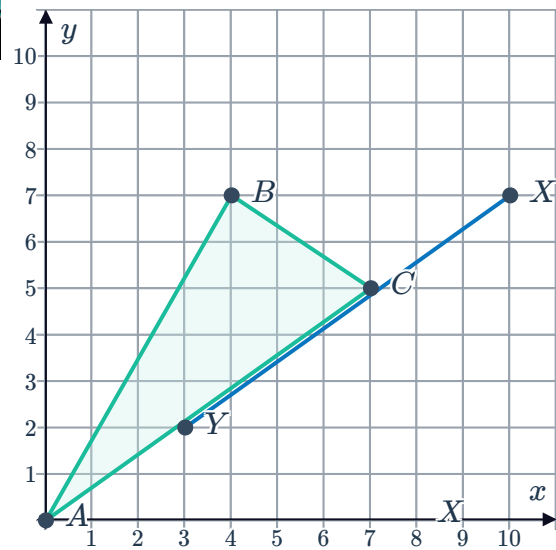


2 Consider the following diagram.

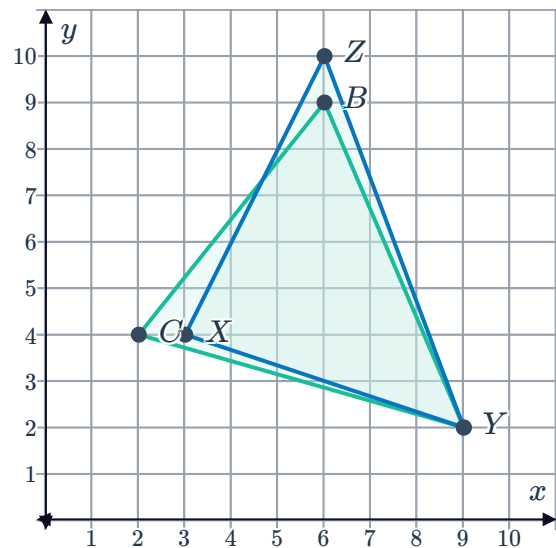
- a Describe the translations that must be applied to move X to A .
- b Hence, which points of $\triangle ABC$ will Y and Z translate to?
- c Are $\triangle ABC$ and $\triangle XYZ$ congruent?



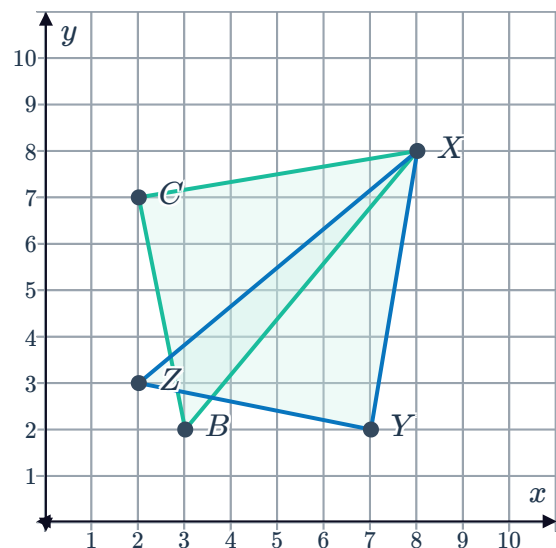
- 3 In the diagram shown, X is a translation of C , and Y is a translation of A . Plot Z on the given coordinate plane so that the resulting $\triangle YZX$ is congruent to $\triangle ABC$.



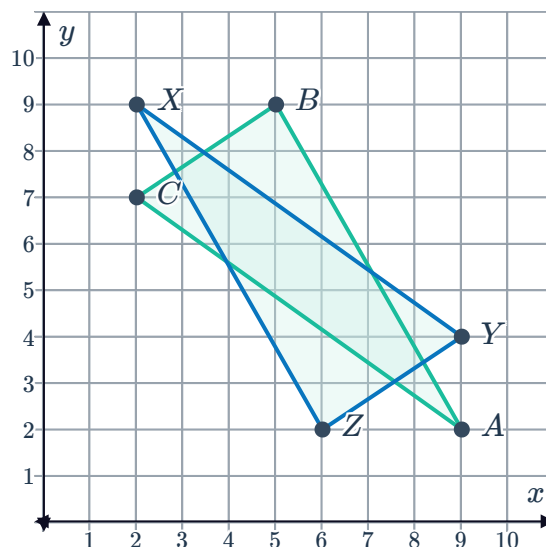
- 4 Consider the two triangles drawn in the following diagram.
Are $\triangle YBC$ and $\triangle YXZ$ congruent?



- 5 Consider the two triangles drawn in the diagram shown.
Are $\triangle XBC$ and $\triangle XYZ$ congruent? Explain your answer.



- 6 Consider the two triangles drawn in the diagram below.
Are $\triangle ABC$ and $\triangle XZY$ congruent? Explain your answer.

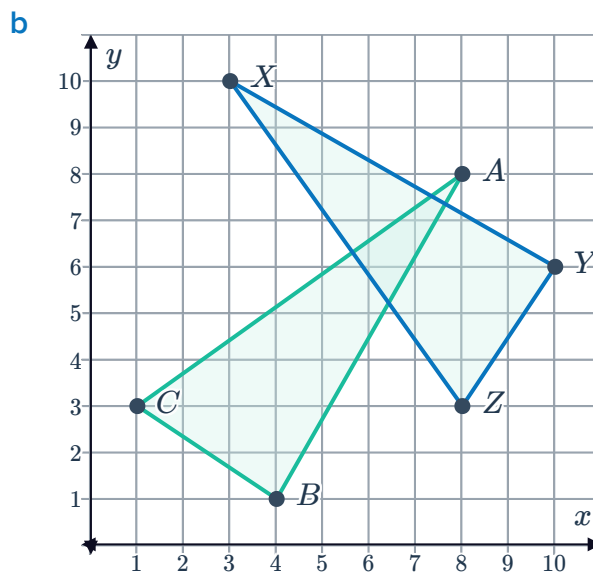
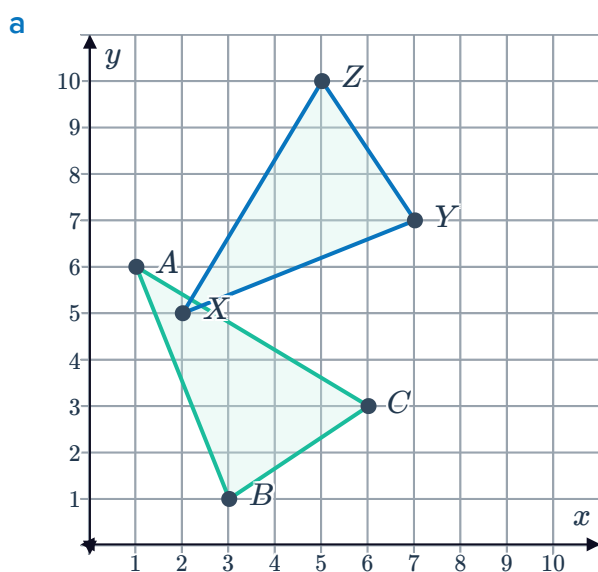


- 7 For each of the following diagrams:

- Find AB .
- Using the same method as the previous part, or otherwise, complete the following table of segment lengths.

Segment	$\overline{AB}$	$\overline{BC}$	$\overline{CA}$	$\overline{XY}$	$\overline{YZ}$	$\overline{ZX}$
Length						

- Are $\triangle ABC$ and $\triangle XYZ$ congruent?



c

